

Thesis Project
ANNA SOFIA MORO

Analysis of H8 bent-crystal measurements

121st BE-ABP-NDC section meeting 26.08.2026

WHAT IS 'CHANNELING' IN CRYSTALS?

THE PHYSICS

- *Channeling effect*

Charged particles get trapped and guided between regular atomic planes

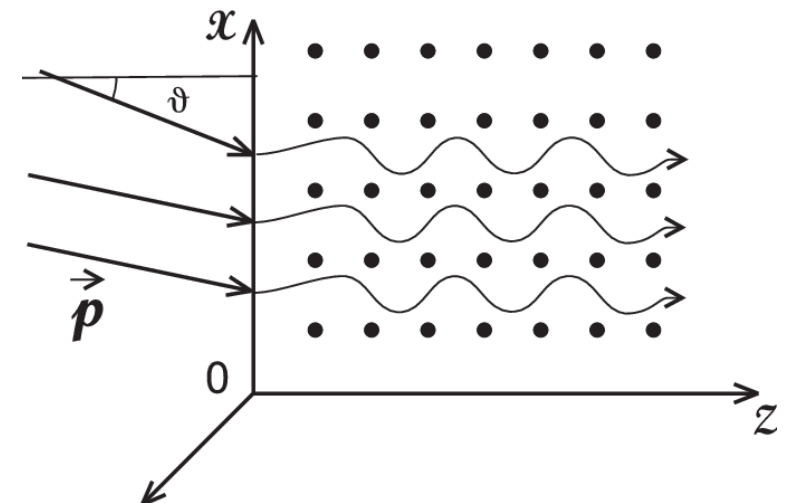
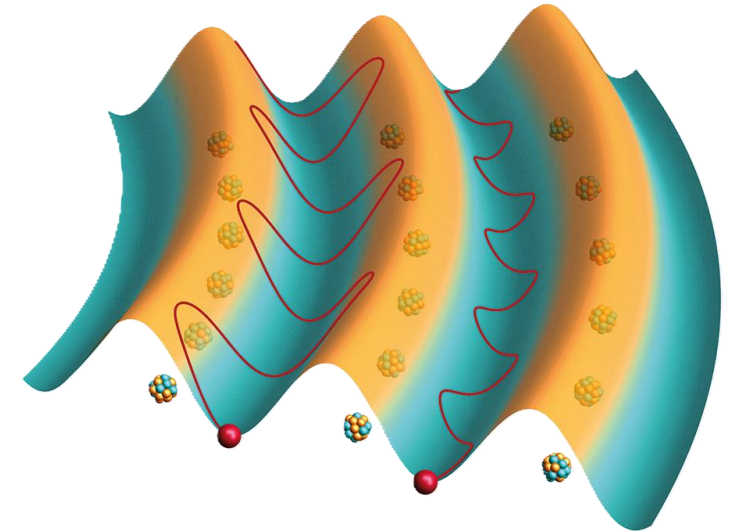
- *Beam deflection*

Bending the crystal physically steers the beam's trajectory

- *Advantage*

A compact alternative to traditional magnets for deflecting or extracting beams

Entry condition: the Lindhard angle θ_L
incident angle $< \pm 1/2 \theta_L$
 $\approx 13 \mu\text{rad}$ @ 180 GeV



THE TWOCRYST EXPERIMENT

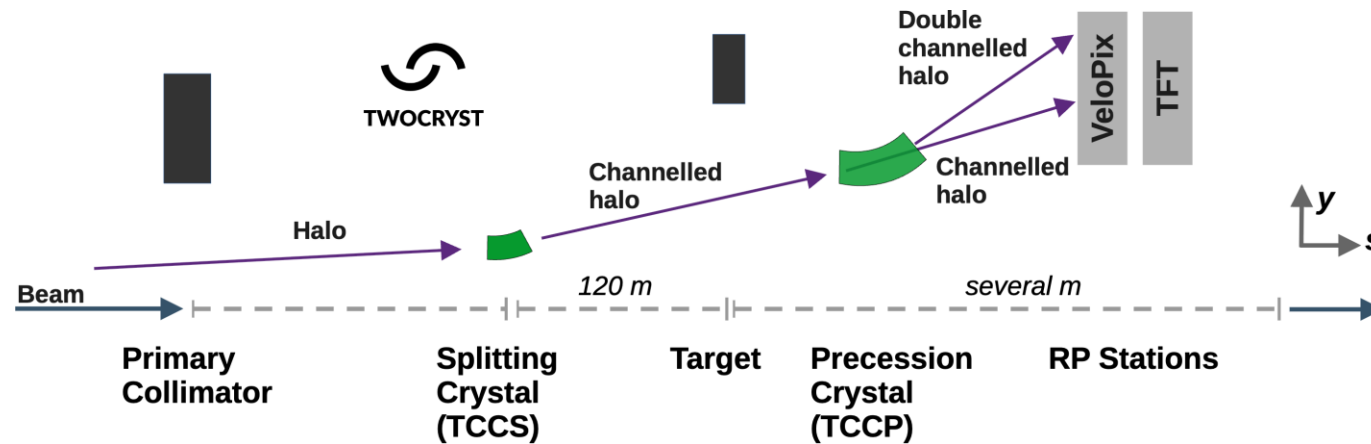
THE PROJECT

before

Bent crystals used in collimation systems to deflect particles and for beam extraction

(2025 - now)

Validate the **operational feasibility, crystal performance**, required for a double-crystal fixed-target experiment inside the LHC (*TeV scales*)



future developments

ALADDIN: exploits the strong electromagnetic fields inside bent silicon crystals to steer the Λ_c^+ particles to induce a strong deflection (spin precession) and measure their **dipole moment**

①

DATA

2024 dataset for two long crystals

Taken at SPS-H8 test bench with 180 GeV beam (70% protons and 30% pions)

②

GOAL OF MY WORK

Measure the **channeling efficiency** ϵ_{ch} and the crystal torsion on newly fabricated long crystals.

③

WHAT IS NEW

These crystals were fabricated directly by CERN, unlike the ones mounted on TWOCRIST.

MY ANALYSIS

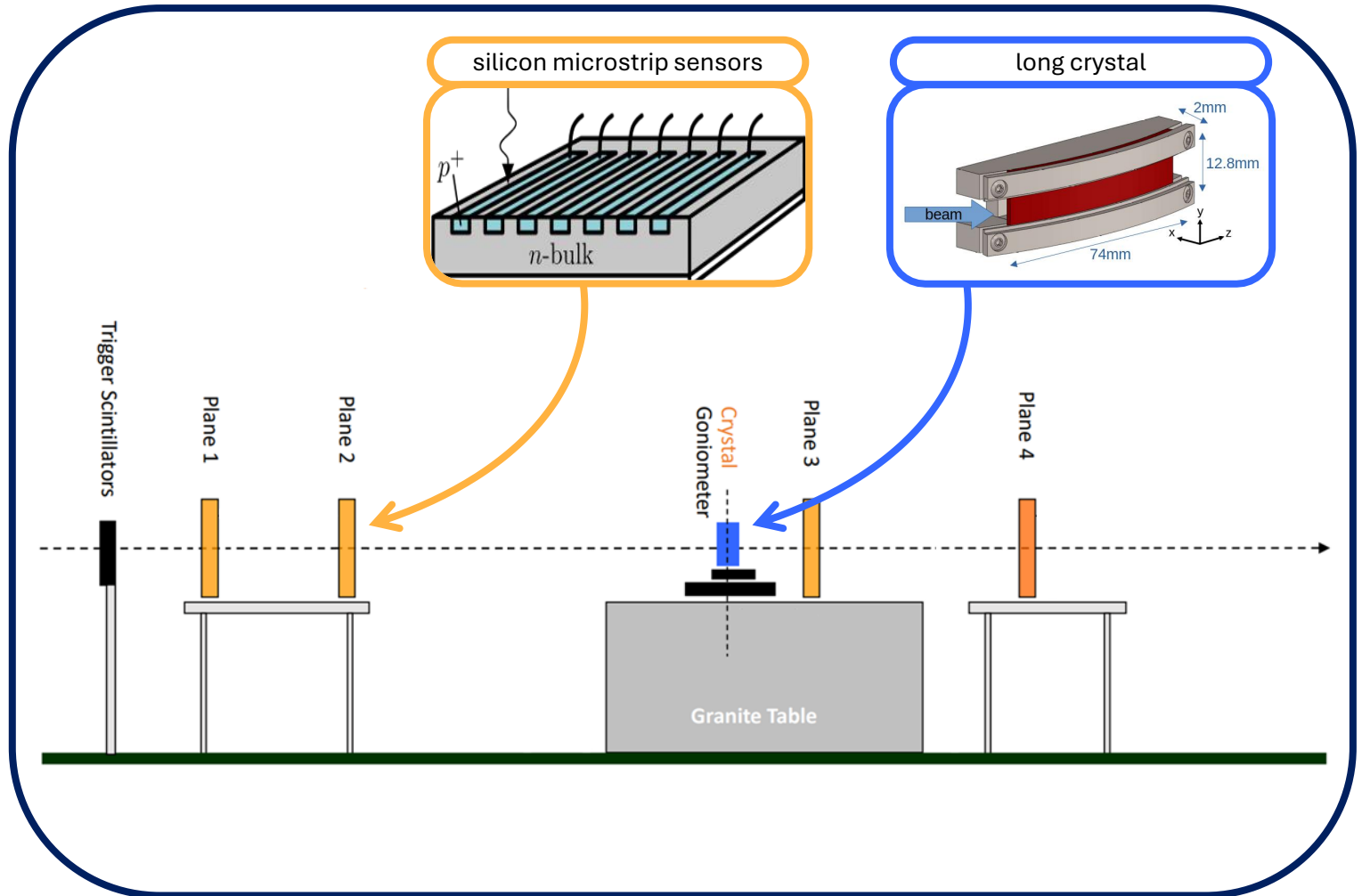
EXPERIMENTAL SETUP

1

DATA

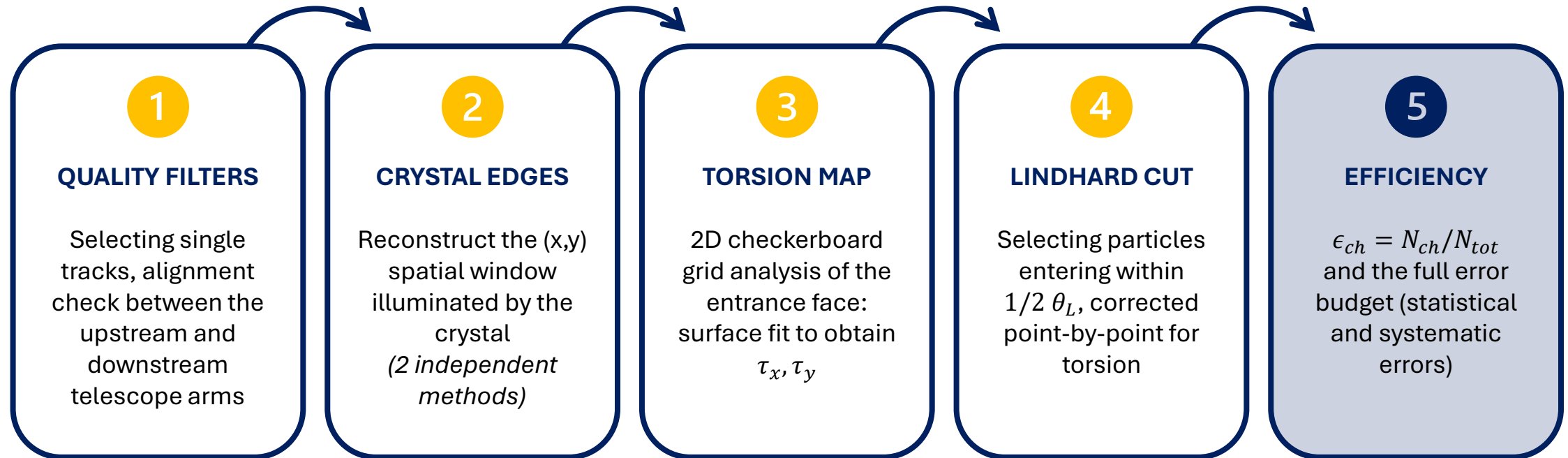
2024 dataset for two long crystals

Taken at SPS-H8 test bench with 180 GeV beam (70% protons and 30% pions)



THE ANALYSIS PIPELINE

METHOD



Analysis implemented in Python on RDataFrame (ROOT).

ANALYSIS: DETECTOR ALIGNMENT

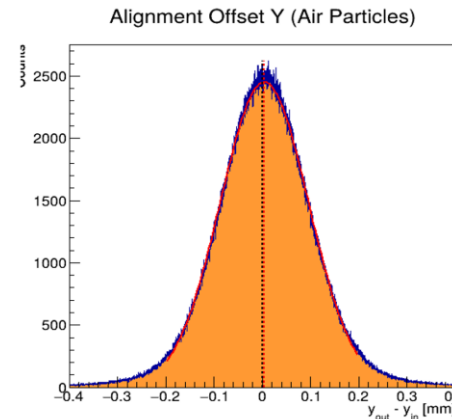
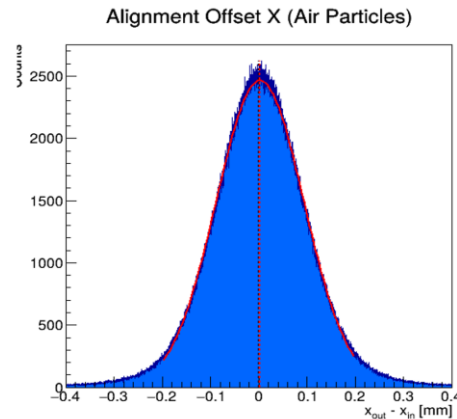
PRELIMINARY ANALYSIS

To ensure that the upstream and downstream detectors are aligned, we check the deviation of particles that did not hit the crystal.

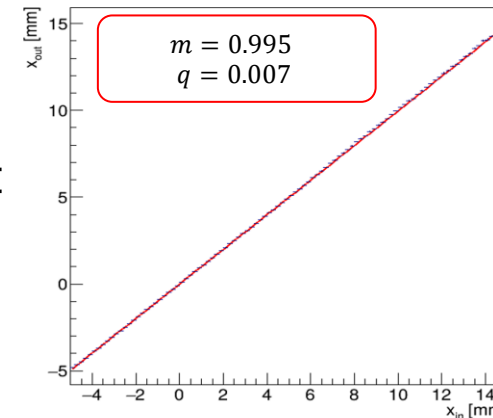
1

QUALITY FILTERS

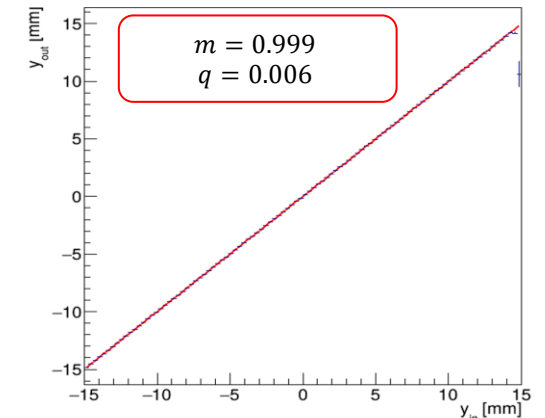
Selecting single tracks, alignment check between the upstream and downstream telescope arms



x_{out} vs x_{in} (air only)



y_{out} vs y_{in} (air only)



We can also check the alignment by doing a linear fit between the upstream and downstream profiles.

ANALYSIS: CRYSTAL EDGES

PRELIMINARY ANALYSIS

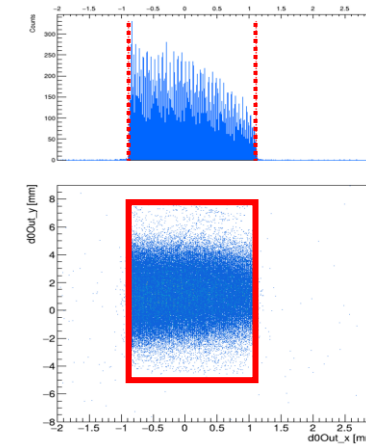
2

CRYSTAL EDGES

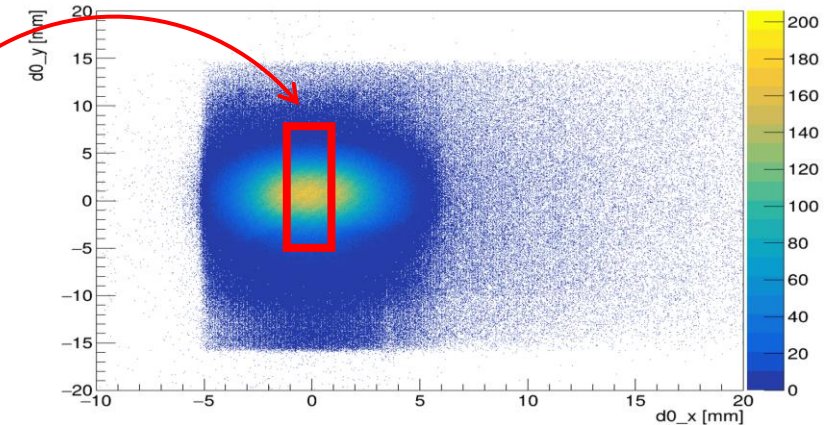
Reconstruct the (x,y)
spatial window
illuminated by the
crystal
(2 independent
methods)

METHOD 1: CHanneled PARTICLES

- Isolate already channeled particles (only uses $\sim 20\%$ of data)
- Their impact position (x, y) traces the crystal footprint
- Use of a sliding window method, which maximize the number of particles inside
- To project the downstream edges back upstream a shift is needed (due to the lateral geometric drift along the x -axis)



Selected window
 $2 \text{ mm} \times 12.8 \text{ mm}$



METHOD 2: MULTIPLE COULOMB SCATTERING

ANALYSIS: CRYSTAL EDGES

PRELIMINARY ANALYSIS

2

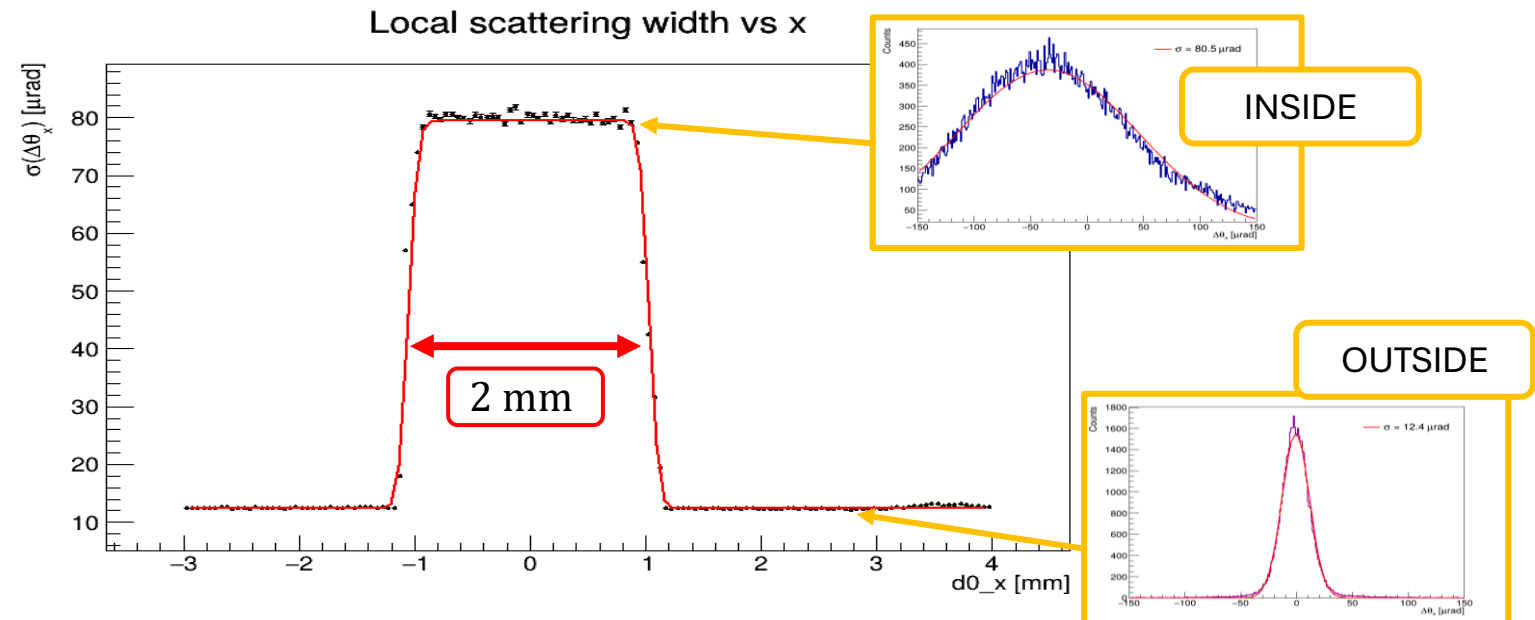
CRYSTAL EDGES

Reconstruct the (x,y)
spatial window
illuminated by the
crystal
(2 independent
methods)

METHOD 1: CHANNELED PARTICLES

METHOD 2: MULTIPLE COULOMB SCATTERING

- Particles that physically crosses the crystal picks up extra Coulomb scattering, the ones outside the crystal do not
- The width σ of the outgoing angular distribution is measured
- A sharp transition marks the edges



ANALYSIS: MAPPING THE TORSION

PRELIMINARY ANALYSIS

- The torsion effect is not negligible in long crystals (> 1 cm)
- The crystal is not perfectly rigid: the orientation of the crystalline planes rotates slightly across its surface
- The optimal entry angle is therefore not unique, but varies point by point $\theta_0(x, y)$

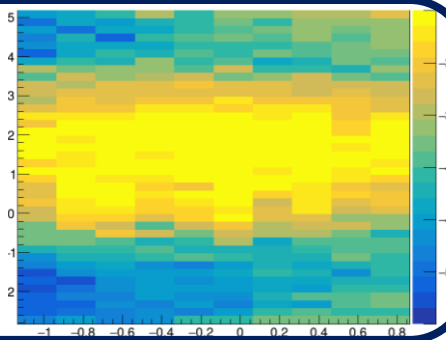
3

TORSION MAP

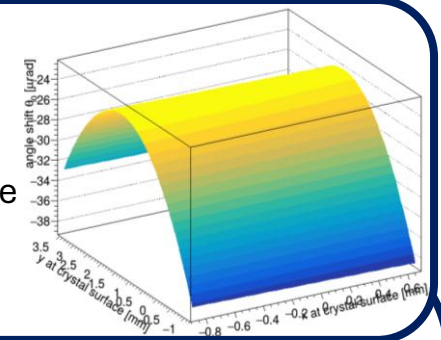
2D checkerboard grid analysis of the entrance face:
surface fit to obtain
 τ_x, τ_y

Beam spot divided
into a 0.1 mm (x,y)
grid

Angular scan
for each cell
 $\rightarrow$ local θ_0
that maximize
efficiency



Surface fit
(parabolic)
to extract the
global τ_x, τ_y



RESULTS:

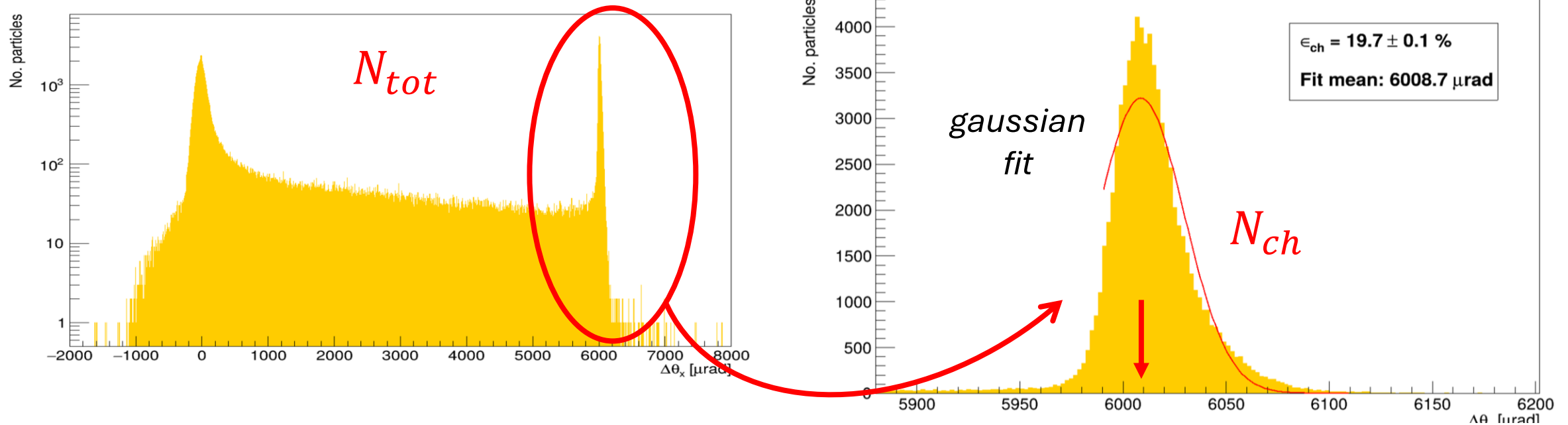
Torsion along the y -axis: $\tau_y = 5.98(3) \mu\text{rad/mm}$
Negligible torsion along the x -axis: $\tau_x = 0.1(1) \mu\text{rad/mm}$

CHANNELING EFFICIENCY

ANALYSIS

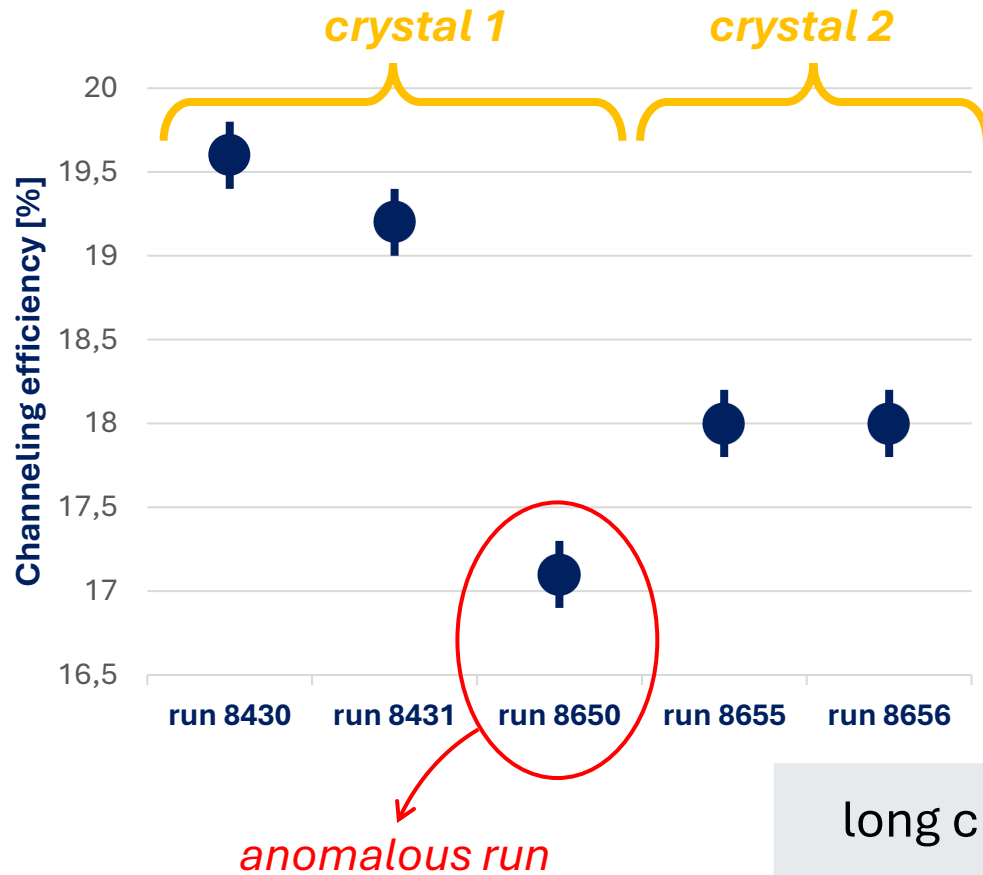
CHANNELING EFFICIENCY

$$\epsilon_{ch} = \frac{N_{ch}}{N_{tot}}$$



CHANNELING EFFICIENCY: RESULTS

ANALYSIS



RUN 8430 (REFERENCE)

$(19.6 \pm 0.1[\text{stat}] \pm 0.2[\text{syst}])\%$

	ϵ_{ch}	τ_y
long crystal 1	$(19.4 \pm 0.1 \pm 0.2)\%$	$(5.98 \pm 0.03)\mu\text{rad/mm}$
long crystal 2	$(18.0 \pm 0.1 \pm 0.2)\%$	$(3.77 \pm 0.01)\mu\text{rad/mm}$

CHANNELING EFFICIENCY: RESULTS

ANALYSIS

Long crystal 1

$(19.4 \pm 0.1[\text{stat}] \pm 0.2[\text{syst}])\%$

Long crystal 2

$(18.0 \pm 0.1[\text{stat}] \pm 0.2[\text{syst}])\%$

ERROR BUDGET

Statistical error:

- Binomial error
- Fit errors

0.1%

Systematic error:

- Fit model choice
- Binning choices

0.2%

CHANNELING EFFICIENCY: RESULTS

ANALYSIS

long crystal 1	$(19.4 \pm 0.1 \pm 0.2)\%$	$(5.98 \pm 0.03)\mu\text{rad/mm}$
long crystal 2	$(18.0 \pm 0.1 \pm 0.2)\%$	$(3.77 \pm 0.01)\mu\text{rad/mm}$
	ϵ_{ch}	τ_y



On article *Bandiera et al.* (2025):

It was found and efficiency of $\epsilon_{ch} = (15.8 \pm 0.1)\%$ and a torsion of $\tau_y = (24.4 \pm 0.2)\mu\text{rad/mm}$ for the TCCP long crystal.

Instead our two crystals can reach a maximum efficiency of 23 – 25% in some areas.

WHAT'S NEXT: SIMULATION

NEXT STEPS



Set up a **Geant4 (BDSIM)** simulation of the long crystals.



The actual data serve as a **benchmark**.



The simulation reproduces the channeling physics without imperfections (**higher** simulated efficiency).

→ Multiple Coulomb scattering will be added to simulated results based on empirical data, to benchmark simulation model.



The gap between the simulation and the data serves as a quantitative measure of the impact of torsion and other non-idealities.

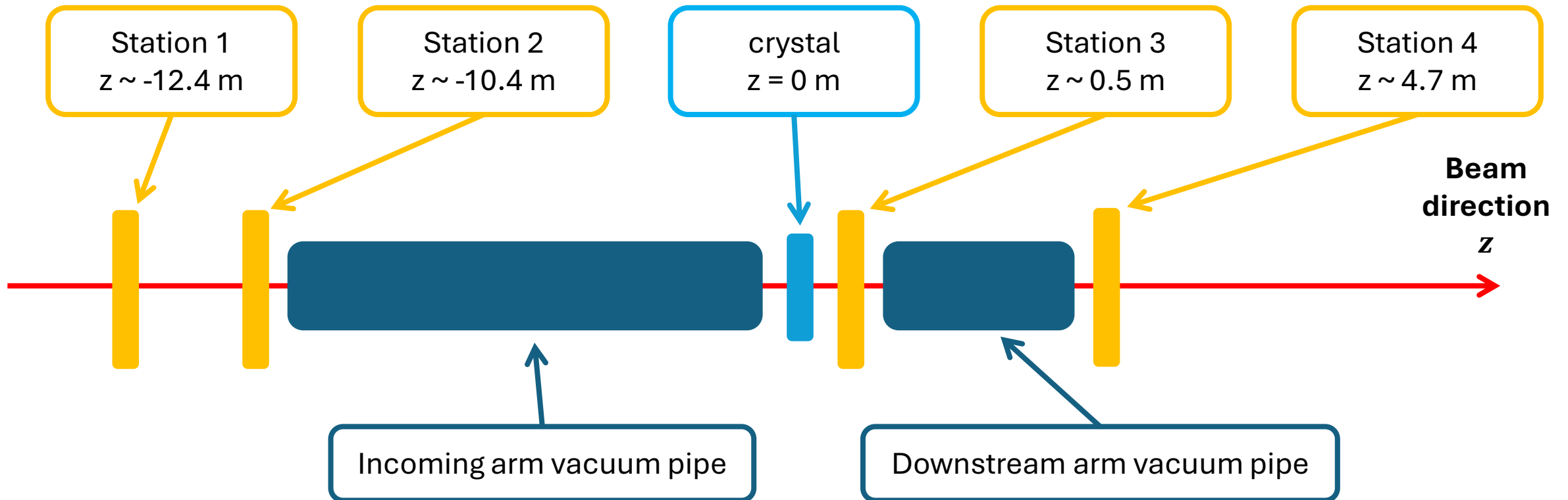
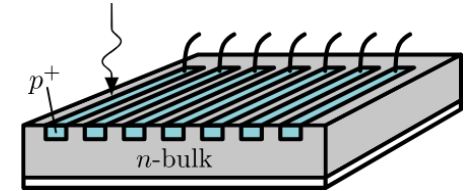
BACKUP SLIDES

SCHEMATIC LAYOUT

BACKUP SLIDE

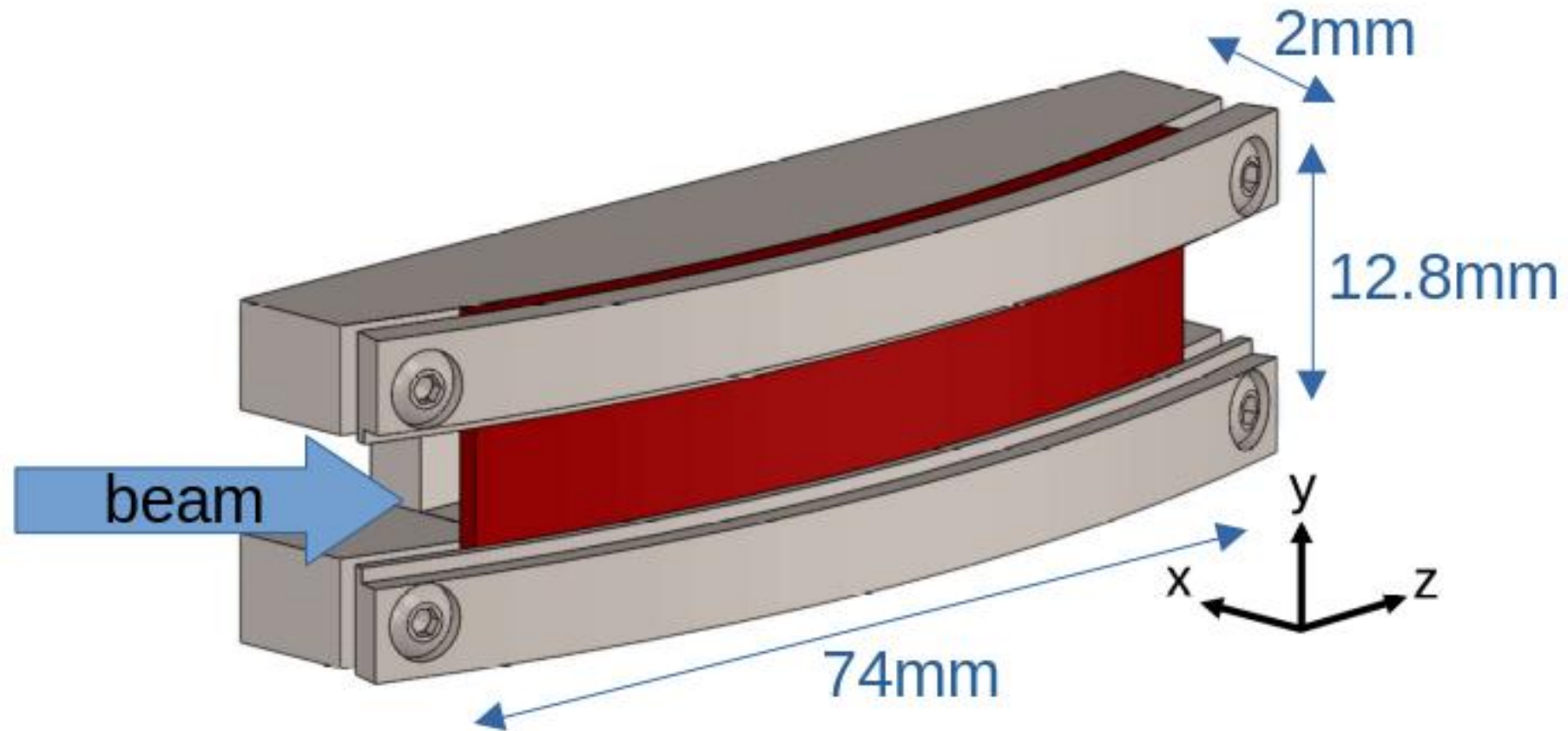
Beam: H8 secondary beamline off the SPS, driven by 400 GeV protons
Spectrometer selection 180 GeV hadrons (30% π^+ and 70% p)

Trackers:
2 silicon microstrip sensors
639 strips, 98 mm length with a ~ 7 μm resolution



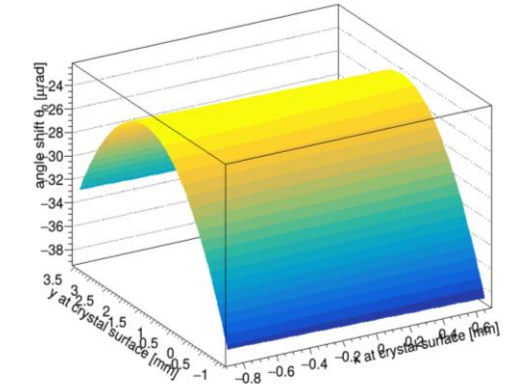
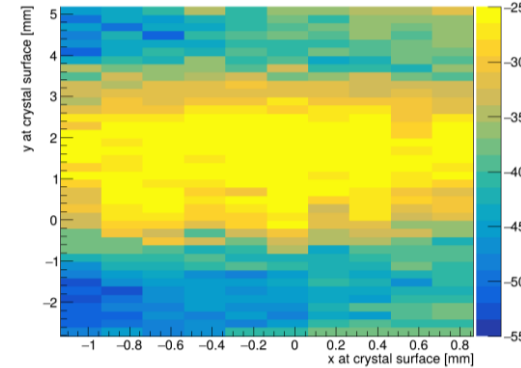
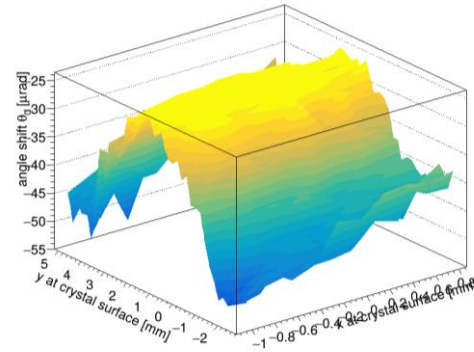
SCHEMATIC LAYOUT

BACKUP SLIDE



TORSION FIT

BACKUP SLIDE



The 2d map is fitted through a parabolic function:

$$\theta_{corr}(x, y) = \theta_0 + \tau_x \cdot x + \tau_y \cdot y + p_3 y^2$$

Now we correct the effect of torsion using the parameter $\theta_0(x, y)$ found for each bin:

instead of the Lindhard cut $|\theta_{in,x}| \leq \frac{1}{2} \theta_L$ we will use

$$\left| \theta_x^{in} - \left(\theta_x^{off} + \tau_y \cdot d_y + \tau_x \cdot d_x \right) \right| \leq \frac{1}{2} \theta_L$$

ANOMALOUS RUN 8650

BACKUP SLIDE

Due to an issue, the run was taken at 150 GeV instead of 180 GeV.

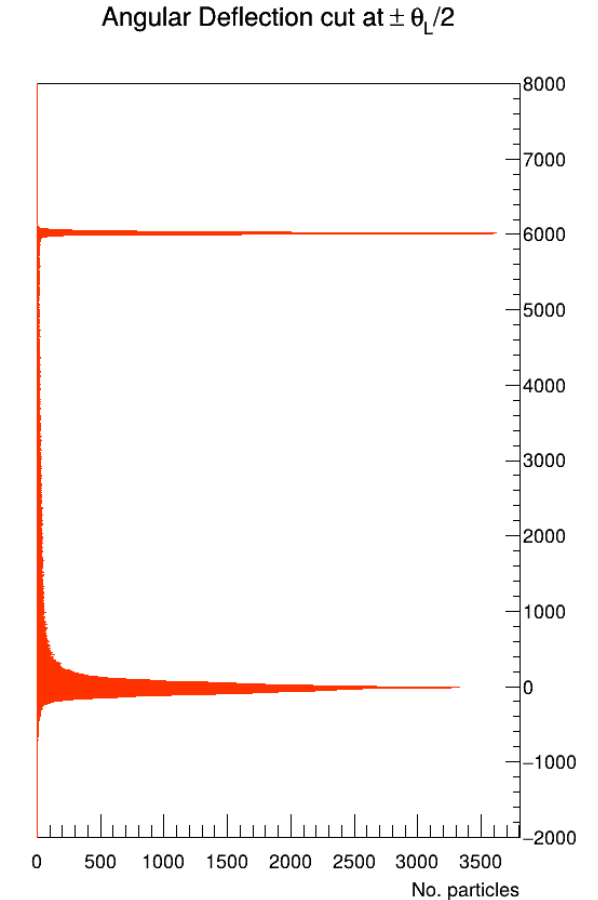
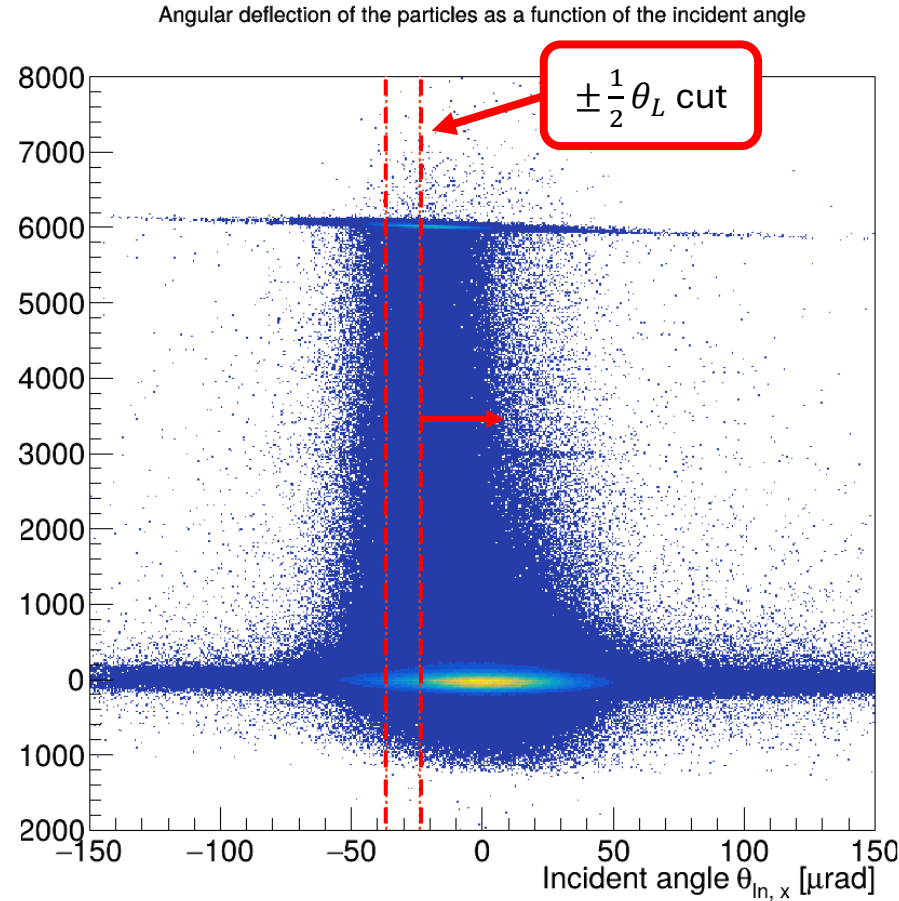
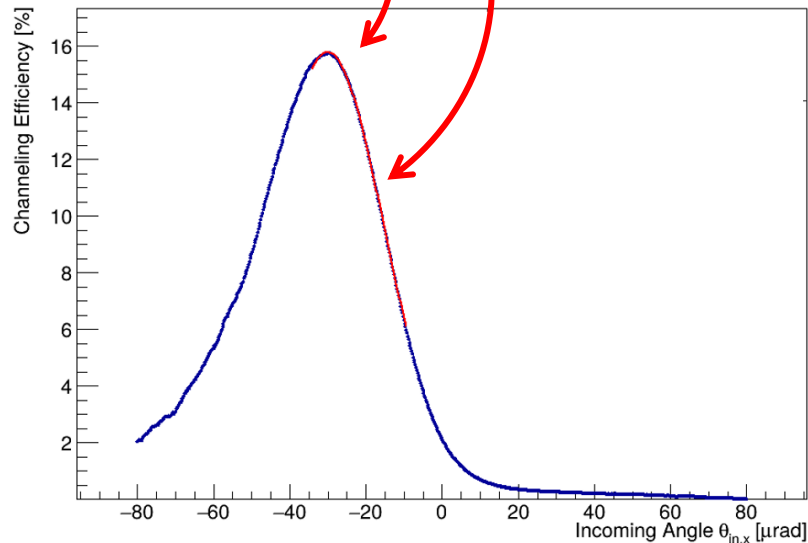
- Wider channeling peak
 - More multiple scattering ($\propto 1/p$)
- Lower efficiency
 - De-channeling more likely at lower energies
- Larger critical Lindhard angle

ANGLE SELECTION

BACKUP SLIDE

For each square bin:

- variation of θ_0
- select particles within $\leq \frac{1}{2} \theta_L$
- compute ϵ_{ch}
- select θ_0 that maximizes ϵ_{ch} (through gaussian fit)



EFFICIENCY CURVE

BACKUP SLIDE

Crystal efficiency as a function of the incoming angle, after application of the torsion correction. The value of the peak is the efficiency of the crystal when it is perfectly aligned to the beam.

